CHEMICAL REACTIONS AND EQUATIONS

1. State any three observations which help to determine whether a chemical reaction has taken place. Write an activity which may justify at least two of the listed observations.

2. Mention three ways by which a usual chemical equation can be made more informative.

3. Consider the following reaction:

$$Pb(NO_3)_{2(s)} \xrightarrow{Heat} PbO_{(s)} + NO_{2(g)} + O_{2(g)}$$

(a) Name the gases produced in the above reaction.
(b) Balance the above chemical equation.
(c) Name the type of chemical reaction.

4. (a) When hydrogen gas burns in presence of oxygen, water is formed and when water is electrolyzed then hydrogen and oxygen gases are produced. State the kind of reaction that takes place
(i) in the first case
(ii) in the second case
(b) In the experimental set-up for electrolysis of water, hydrogen, and oxygen gases are produced at the cathode and anode respectively. Mention the ratio of the volumes of hydrogen and oxygen gases.

5. Food when left for a long time tastes/smells bad. What is this condition called? Mention any four ways by which we can prevent this condition.